

Midterm 3: Persistence/Distribution/Virtual Machines

Time: 120 minutes

Points: 75

Questions: 75

Instructions

This exam is a closed book, closed notes. When you begin the exam, you will need to agree with the following statements: I understand that this exam must be worked on independently. I will not communicate or collaborate with any others during the exam in any way, for example:

- *I will not chat, text, or post any questions or information*
- *I will not ask for, receive, or provide help to others*
- *I will notify the instructor if I am aware of others violating this policy*

Unless stated (or implied) otherwise, you should make the following assumptions:

- *The OS manages a multiprocessor (multi-core) that can run multiple threads simultaneously*
- *Leading zeros can be removed from numbers (e.g., 0x06 == 0x6)*
- *Hex numbers are represented with a preceding "0x"*
- *All the code mentioned in the paper is assumed to compile properly. You can ignore any syntax error if there is any.*

You have 120 minutes to complete this exam.

Section - 1: True/False. There are 35 questions, and each question is worth 1 point.

1. When interacting with a device, interrupts reduce the latency of completing an I/O operation compared to polling.
 - a. True
 - b. False
2. Every device has a device driver
 - a. True
 - b. False
3. A network device driver can talk to any kind of network card
 - a. True
 - b. False
4. Process B can use the CPU when process A is waiting for an IO operation to complete.
 - a. True
 - b. False
5. SSTF is more fair than FCFS
 - a. True
 - b. False
6. SPTF scheduling over a set of N requests will order the requests resulting in the minimum possible total positioning time.
 - a. True
 - b. False
7. On an HDD (assume no on-disk caching), a workload of sequential writes will obtain significantly higher throughput than a workload of random reads.
 - a. True
 - b. False
8. When accessing an HDD, the OS knows the track number where each sector is located.
 - a. True
 - b. False
9. To obtain peak sequential write bandwidth on a RAID-5 array, the contents of the parity block are calculated by performing the following steps – reading the old data, reading the old parity, writing the new data, and writing the new parity.
 - a. True
 - b. False

10. To recover lost blocks from a failed disk in a RAID-5 array, one must know which blocks hold data and parity.
- True
 - False
11. With RAID-1, the steady-state throughput of random reads and random writes are identical.
- True
 - False
12. RAID-1 can deliver the correct data for a read operation if one disk in the array silently returns incorrect data.
- True
 - False
13. In LFS, when an inode is written to a segment, all the data blocks that this inode points to are also written to the same segment.
- True
 - False
14. In LFS, given an inode number, reading an inode requires at most 1 disk read.
- True
 - False
15. In FFS, an inode may point to a data block in another cylinder group.
- True
 - False
16. To reduce seek costs, FFS generally tries to keep directory inodes and directory data blocks in the same cylinder group.
- True
 - False
17. In LFS, when an inode is written to a new segment on disk, any directory entries that refer to that inode are also written to the new segment.
- True
 - False
18. In an ordered meta-data journaling file system, data blocks must be written to their in-place locations on the disk before the corresponding transaction is committed.
- True
 - False

19. A file system that uses contiguous allocation for files is likely to suffer from more external fragmentation than one that uses extents.
- a. True
 - b. False
20. In Unix-based file systems, multiple file names can map to the same inode number.
- a. True
 - b. False
21. In Unix-based file systems, directory entries contain each file's size and permission information.
- a. True
 - b. False
22. In Unix-based file systems, the write () system call returns to the user when the new data and the changed inode have been persisted to disk.
- a. True
 - b. False
23. An inconsistency is found if FSCK finds a valid inode referenced by two different directory entries.
- a. True
 - b. False
24. In a Unix-style file system, creating a new (empty) file F in an existing directory D involves writing exactly these 3 disk blocks: inode D, data for D, the inode for F.
- a. True
 - b. False
25. In Unix-based file systems, two processes may share the same file object.
- a. True
 - b. False
26. With TCP, when a timeout expires before an ACK packet is received, it means the receiver did not get the sender's message.
- a. True
 - b. False
27. With RPC, the wrappers on the client are responsible for sending and receiving messages to the server.
- a. True
 - b. False

28. When using RPC, the request message needs an ACK so the client knows it was successfully delivered to the server, and the response needs an ACK so the server knows it was received by the client
- a. True
 - b. False
29. A programmer can replace local function calls with calls to RPC client wrappers without any other change to their code
- a. True
 - b. False
30. In a distributed system, the sender of a message knows whether the message was received by its intended recipient
- a. True
 - b. False
31. In a distributed system, a sender knows if a recipient server has crashed or not
- a. True
 - b. False
32. Idempotent operations will still give the correct results if they are not executed by the NFS server since the NFS server is stateless.
- a. True
 - b. False
33. In NFSv2, the NFS server tracks open file descriptors to avoid performing a path name traversal on every file read or write.
- a. True
 - b. False
34. An inode structure typically contains a field indicating the name of this file.
- a. True
 - b. False
35. File systems using File allocation tables (FAT) have poor sequential performance
- a. True
 - b. False

Section - 2: Multiple Choice. There are 25 questions and each is worth 1 point

36. A hard disk spins at 6000 RPM. What is the **average rotational delay**?
- a. 1 ms
 - b. 5 ms
 - c. 10 ms
 - d. 20 ms
 - e. 40 ms
37. A hard disk has a track size of 8 MB and spins at 6000 RPM. What is the transfer speed?
- a. 8 MB/sec
 - b. 160 MB/sec
 - c. 800 MB/sec
 - d. 1200 MB/sec
38. For random read operations with the same number of disks
- a. RAID 5 has better throughput than RAID 1 + RAID 0
 - b. RAID 5 has worse throughput than RAID 1 + RAID 0
 - c. RAID 5 has the same throughput as RAID 1 + RAID 0
39. For random write operations with the same number of disks
- a. RAID 5 has better throughput than RAID 1 + RAID 0
 - b. RAID 5 has worse throughput than RAID 1 + RAID 0
 - c. RAID 5 has the same throughput as RAID 1 + RAID 0
40. You are storing a 100MB file and your block size is 4KB. You have 5 disks to store your data. Suppose you use JBOD - just a bunch of disks, each a separate volume/file system. What is your **sequential throughput for a single file** compared to a single disk?
- a. 1 * single disk
 - b. 2 * single disk
 - c. 4 * single disk
 - d. 5 * single disk
41. You are storing a 100MB file and your block size is 4KB. You have 5 disks to store your data. Suppose you use RAID5. What is your **sequential read throughput for a single file** compared to a single disk?
- a. 1 * single disk
 - b. 2 * single disk
 - c. 4 * single disk
 - d. 5 * single disk
42. In an FFS-like file system
- a. multiple inodes can point to the same file contents
 - b. Multiple directory entries can point to the same file inode
 - c. Multiple inodes can point to the same directory entry
 - d. Multiple inodes can point to the same file descriptor

43. In an FFS-like file system, you have read the superblock but nothing else and now want to append to the file `"/Users/swift/537-grades.txt"`. Assume a 4kb block size, inodes have 12 direct pointers, 1 indirect pointer, and 1 double indirect pointer, and block pointers are 4 bytes. If the file is 64KB and you are appending 100 bytes, how many blocks do you read and write
- a. 7 read + 2 write
 - b. 8 read + 3 write
 - c. 8 read + 4 write
 - d. 9 read + 3 write
 - e. 9 read + 4 write
44. Consider an inode structure that comprises 8 direct blocks, 1 single-indirect block, 1 double-indirect block, and 1-triple indirect block. Assume the block size is 4 KB and 4-byte pointers. What is the largest size file using only the direct blocks?
- a. 8 KB
 - b. 16 KB
 - c. 32 KB
 - d. 128 KB
45. Consider an inode structure that comprises 8 direct blocks, 1 single-indirect block, 1 double-indirect block, and 1-triple indirect block. Assume the block size is 4 KB and 4-byte pointers. What is the largest size file that uses only the direct blocks and 1 single-indirect block?
- a. 4 KB
 - b. 36 KB
 - c. 4096 KB
 - d. 4128 KB
46. LFS requires garbage collection/cleaning to make empty segments. Suppose you have segments that are 80% filled with live data, and that disk reads and writes take equal time. Compared to the time of writing to an empty segment, how long does it take to make an empty (clean) segment then write to it?
- a. 1X
 - b. 3X
 - c. 5X
 - d. 7X
 - e. 9X
47. An LFS file system can choose to clean segments continuously, or wait until the file system runs out of clean segments before cleaning a segment. In which case will it do more work (reading/writing more data) to create clean segments?
- a. Cleaning continuously
 - b. Cleaning only when out of clean segments

48. Early/simple SSDs use a direct map strategy for mapping logical block numbers to flash pages. Which workloads perform poorly for a direct map strategy:
- Random read
 - Random write
 - Sequential read
 - Sequential write
49. Suppose you have an SSD with 128KB erase blocks and 4KB pages. The SSD can do 10,000 writes per second, and 500 erases per second. Using a direct map strategy for mapping logical blocks to flash pages, what is (approximately) the random throughput for 4kb writes in operations per second?
- 250
 - 500
 - 1000
 - 10,000
50. Suppose you have an SSD with 128KB erase blocks and 4KB pages. The SSD can do 10,000 writes per second, and 500 erases per second. Using a logging strategy for mapping logical blocks to flash pages, what is the random throughput for 4kb writes in operations per second?
- 250
 - 500
 - 1000
 - 10,000
51. With UDP, which of these failures cannot happen
- Packet is dropped
 - Packet is corrupted
 - Packet is delivered twice
 - Packets are delivered out of order
52. After sending a TCP message, the sender is guaranteed that:
- The receiver gets at most one copy of the message
 - The receiver gets at least one copy of the message
 - The receiver gets exactly one copy of the message
53. You open a file from one process and read from it. Another process on the same machine calls `unlink()` on the same file. Your process then reads from the file again. What happens?
- The read completes successfully
 - The read fails because the file has been deleted
 - The `unlink` fails because the file is open

54. You open a file stored on an NFS file server from one process and read from it. Another process on a different machine calls `unlink()` on the same file. Your process then seeks to a new location much further into the file reads from the file again. What happens?
- a. The read completes successfully
 - b. The read fails because the file has been deleted
 - c. The unlink fails because the file is open
55. You open a file on an NFS server. A process on a different machine renames the file and moves it to a different directory. You then seek further into the file and read again. What happens?
- a. The read completes successfully
 - b. The read fails because the file is no longer there.
56. You open a file on an NFS server. While you have the file open, the NFS server crashes and restarts. You then try to access the file again after the server restarts. What happens?
- a. The access completes successfully
 - b. The access fails because the server restarted
57. When creating a file and then writing to a file, the following blocks are written
- a. File Inode, file data block
 - b. File inode, file data block, directory inode
 - c. File inode, file data block, directory inode, directory data block
 - d. File inode, file data block, data block bitmap, directory inode, directory data block
 - e. File inode, file data block, data block bitmap, directory inode, directory data block, inode bitmap
58. FFS **DOES NOT** place the following things in the same cylinder group
- a. File inode, file data
 - b. Directory entries for a file, file inode
 - c. Directory entries for a directory, directory inode
 - d. File indirect blocks, file data
59. Which workloads perform better on FFS than LFS?
- a. Sequential write followed by sequential read
 - b. Random write followed by random read
 - c. Random write followed by sequential read
 - d. Sequential write followed by random read

60. In LFS, the IMAP is updated
- a. When writing to a file
 - b. When adding a file to a directory
 - c. When cleaning/garbage collecting
 - d. All of the above
 - e. A and B only

Part 3. There are 3 multi-part questions for a total of 13 points.

61. Given this sequence of requests : 68, 74, 47, 51, 29, 44, 38, 66, 95, 87, 77. Suppose the SCAN algorithm handled requests 68 then 74. What will be the next request it processes?
- a. 47
 - b. 66
 - c. 77
 - d. 87
 - e. None of the above
62. Given this sequence of requests : 75, 7, 19, 59, 81, 16, 14, 6, 46, 77. Suppose the CSCAN algorithm handled requests 7, 19, 59, 75, 81 in that order. What will be the next request it processes?
- a. 6
 - b. 14
 - c. 46
 - d. 77
 - e. None of the above
63. Assume the disk head is at track 0, and this sequence of requests arrives: 9, 12, 40, 41, 56, 64, 20, 33. With an FCFS scheduling policy, what is the total seek distance to service these requests?
- a. 55
 - b. 64
 - c. 112
 - d. 121
 - e. None of the above

64. Assume the disk head is at track 0 moving towards higher tracks, and this sequence of requests arrives: 9, 12, 40, 41, 56, 64, 20, 33. With an SCAN scheduling policy, what is the total seek distance to service these requests?
- a. 55
 - b. 64
 - c. 112
 - d. 121
 - e. None of the above
65. Assume the disk head is at track 35, and this sequence of requests arrives: 9, 12, 40, 41, 56, 64, 20, 33. With an SSTF scheduling policy, what is the total seek distance to service these requests?
- a. 55
 - b. 64
 - c. 88
 - d. 93
66. You are storing a 100MB file and your block size is 4KB. You have 4 disks configured as JBOD, and one disk fails
- a. You have lost the complete contents of your file
 - b. You have lost some of the contents of your file
 - c. You have lost none of the contents of your file
 - d. You may or may not have lost the contents of your file
67. You are storing a 100MB file and your block size is 4KB. You have 4 disks configured as RAID0, and one disk fails
- a. You have lost the complete contents of your file
 - b. You have lost some of the contents of your file
 - c. You have lost none of the contents of your file
 - d. You may or may not have lost the contents of your file
68. You are storing a 100MB file and your block size is 4KB. You have 4 disks configured as RAID1 + RAID0, and one disk fails
- a. You have lost the complete contents of your file
 - b. You have lost some of the contents of your file
 - c. You have lost none of the contents of your file
 - d. You may or may not have lost the contents of your file

You have an FFS-like file system that is unlinking (deleting) a file with one data block. The file system must update the inode bitmap, data block bitmap, and directory data block, and directory inode.. Assume the system does not perform any journaling and FSCK is not run after a crash. What happens if a crash occur after only updating the following block(s)?

69. Directory inode

- a. No problem (appears the operation was not performed)
- b. Data/inode leak
- c. Directory points to garbage/incorrect files
- d. File points to garbage
- e. Corrupted directory

70. Inode bitmap

- a. No problem (appears the operation was not performed)
- b. Data/inode leak
- c. Directory points to garbage/incorrect files
- d. File points to garbage
- e. Corrupted directory

71. Data block bitmap

- a. No problem (appears the operation was not performed)
- b. Data/inode leak
- c. Directory points to garbage/incorrect files
- d. File points to garbage
- e. Corrupted directory

72. Directory inode and Directory Data

- a. No problem (appears the operation was not performed)
- b. Data/inode leak
- c. Directory points to garbage/incorrect files
- d. File points to garbage
- e. Corrupted directory

73. Inode bitmap and directory data and directory inode

- a. No problem (appears the operation was not performed)
- b. Data/inode leak
- c. Directory points to garbage/incorrect files
- d. Corrupted directory
- e. File points to garbage

74. How many other students in CS 537 did you know well enough to ask them for help?

- a. 0
- b. 1
- c. 2
- d. 3
- e. 4 or more

75. Should we continue to have study groups in CS 537?

- a. I don't know - I never went to one
- b. I went to at least one and they weren't helpful
- c. I went to a few and/or they were moderately helpful
- d. I went to many and/or found them very helpful